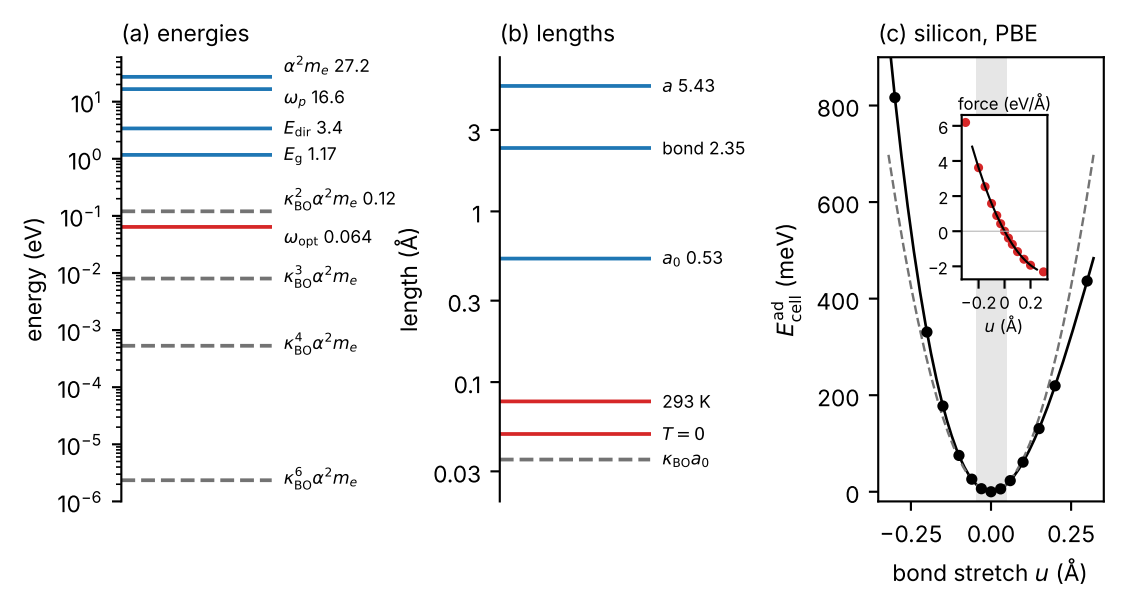
(a) energies
energy (eV)
$\alpha^2 m_e$ 27.2
ω_p 16.6
E_{dir} 3.4
E_{g} 1.17
$\kappa_{\mathrm{BO}}^2 \alpha^2 m_e$ 0.12
ω_{opt} 0.064
$\kappa_{\mathrm{BO}}^3 \alpha^2 m_e$
$\kappa_{\mathrm{BO}}^4 \alpha^2 m_e$
$\kappa_{\mathrm{BO}}^6 \alpha^2 m_e$
(b) lengths
length (Å)
a 5.43
bond 2.35
a_0 0.53
293 K
$T = 0$
$\kappa_{\mathrm{BO}} a_0$
(c) silicon, PBE
$E_{\mathrm{cell}}^{\mathrm{ad}}$ (meV)
force (eV/Å)
u (Å)
bond stretch u (Å)